MGMT 388 Final Project

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The Dataset

The dataset contains 9568 data points collected from a Combined Cycle Power Plant over 6 years (2006-2011), when the power plant was set to work with full load. Features consist of hourly average ambient variables Temperature (T), Ambient Pressure (AP), Relative Humidity (RH) and Exhaust Vacuum (V) to predict the net hourly electrical energy output (EP) of the plant.

Link to the dataset: https://archive.ics.uci.edu/dataset/294/combined+cycle+power+plant

Project Problem

In order to maximize profit from available megawatt hours, it is important to accurately predict full load electrical power output of a base load power plant. Given that hourly full load electrical power output of a combined cycle power plant is influenced most by ambient temperature, atmospheric pressure, relative humidity, and exhaust steam pressure, we can create an accurate predictive regression model given these variables.

Our Analysis & Methodology

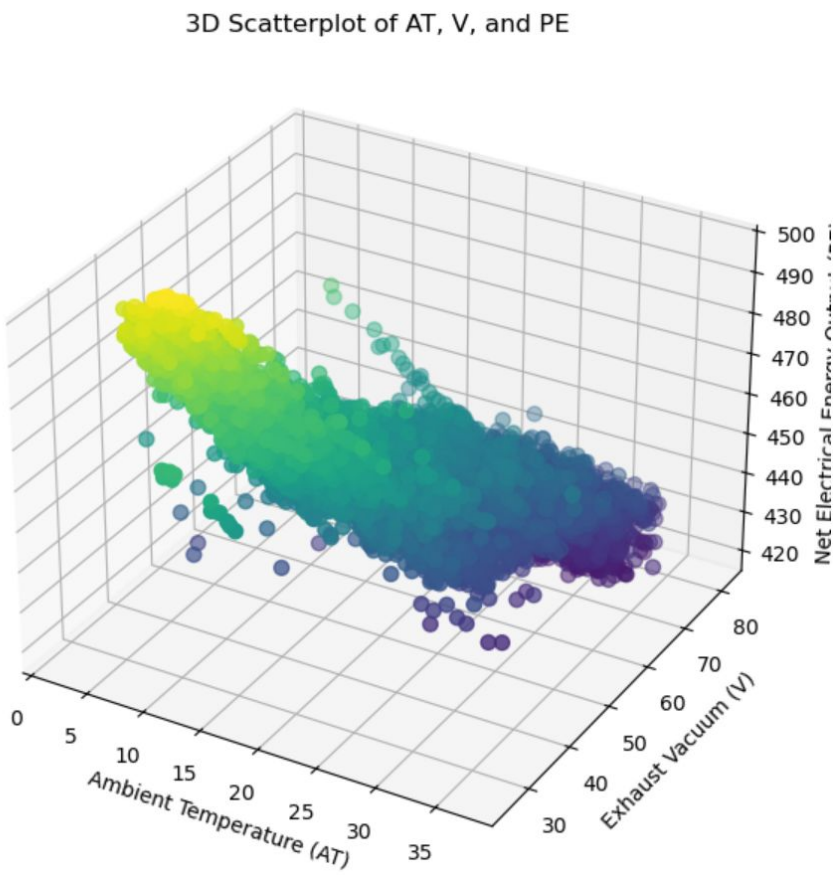
Pre Processing & EDA

01

I loaded the excel file using Pandas and displayed the general information about the data and its descriptive statistics.

I checked for missing values and since there were none, I used the IQR method to filter the dataset and remove all the outliers in each of the columns.

After cleaning the data, exploration showed strong negative correlations for Atmospheric Temperature, Exhaust Vacuum, and Electrical Energy Output.



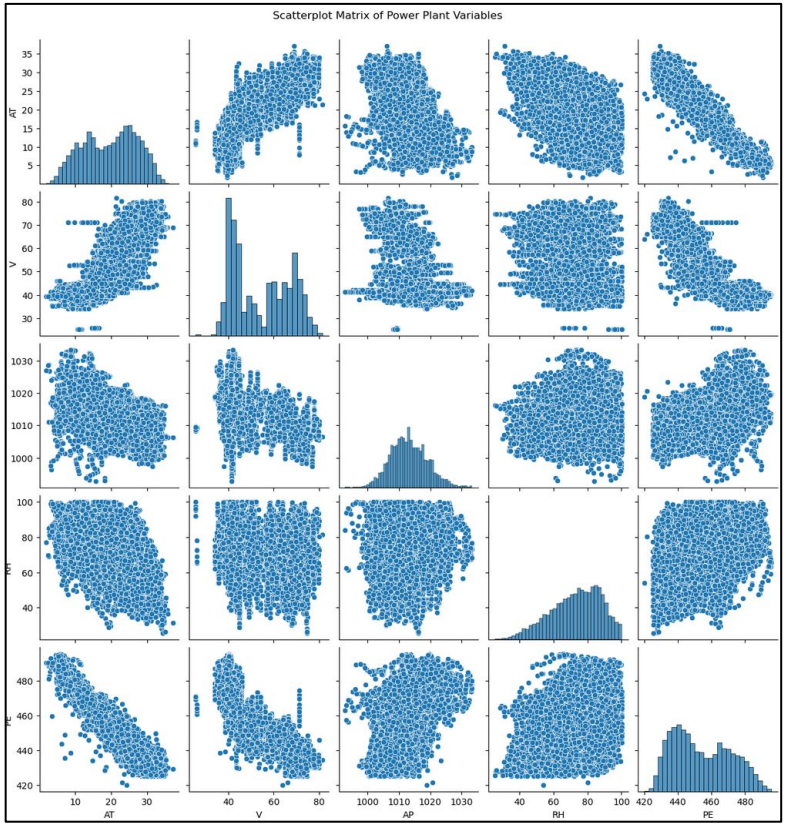
Pre Processing & EDA

02

We created a pariplot for all of the features in the dataset with histograms down the middle for more information. AT, V, PE appear to have bimodal distributions while AP looks normally distributed. From its histogram, RH seems to be fairly left skewed.

Feature Correlations:

- PE has a strong negative correlation with AT and V
- PE has slightly positive correlations with AP and RH
- V and AT have a stronger positive correlation



Feature Engineering & Model Building

I implemented regression analysis to analyze the dataset. Since the 4 independent variables were in different units, I applied the standard scaler method to calculate their z-scores and for future analysis. To find the best regression model. I split the dataset (80% for training, 20% for testing) and building the Linear Regression, Polynomial Regression (degrees = 2, 4, 6), Decision Tree Regression and Random Forest Regression Models.

Model Evaluation and Comparison

After training and testing our dataset using all the regression models, I found that the Random Forest Regression model best fits the data as it resulted in the lowest Mean Squared Error (0.03) and the highest R^2 (0.97).

```
Mean Squared Error for Linear Regression: 0.06775475456682463
Mean Squared Error for Polynomial Regression (degree 2): 0.058076647494414345
Mean Squared Error for Polynomial Regression (degree 4): 0.05432552035199498
Mean Squared Error for Polynomial Regression (degree 6): 0.05103239205257895
Mean Squared Error for Decision Tree Regression: 0.0757436197419307
Mean Squared Error for Random Forest Regression: 0.03494199018996695
```

```
R2 score for Linear Regression: 0.9325315554761302
R2 score for Polynomial Regression (degree 2): 0.942168765946238
R2 score for Polynomial Regression (degree 4): 0.9459040420184242
R2 score for Polynomial Regression (degree 6): 0.9491832546050479
R2 score for Decision Tree Regression: 0.9245764487043432
R2 score for Random Forest Regression: 0.9652056635470465
```

Hyperparameter Tuning and Model Refinement

Using the hyperparameter running methodology for Random Forest Classification from class, we used the random search approach to tune our Random Forest Regression model for the best parameters.

1	Max Depth	Best Value: 31
2	Minimum Samples: Leaf	Best Value: 1
3	Minimum Samples: Split	Best Value: 3
4	n_Estimators	Best Value: 137

Key insights

The Random Forest Regression Model closely predicts the net hourly electrical energy output given the 4 predictors, Average Temperature, Ambient Pressure, Relative Humidity and Exhaust Vacuum, of the plant. Hence, an accurate prediction could generate many benefits such as:

01 | **More Stable Grid:** Electricity demand accurately meets its supply, reducing blackouts (shortage) and wastages (surplus).

02 | **Lower Costs:** Accurate prediction creates optimized operation and lowers operational costs, ensuring economic efficiency.

03 | **More Renewables:** Enables energy companies to integrate green energy (such as solar and wind energy) safely into their systems.

04 | **Higher Profits:** Accurate prediction increases profitability for energy producers by improving market trading and bidding.

05 | **Less Pollution:** Accurate predictions also help to minimize unnecessary emissions and encourages sustainable energy management.

Conclusion

Through various regression ML models, we could identify how accurately the features affect the target variable and the many benefits that the Power Plant would have on various stakeholders such as its employees, environment and the community if it implemented the Random Forest Regression model (most accurate model) to predict its electrical energy output.

Thank you.